

DIY Studio Monitor Full Component Breakdown

Signal Chain & Diagnostics

Signal Chain		Quantity	Item Cost	Total Cost
Scarlett Focusrite	Owned Scarlett Focusrise 4i4	1	\$299.99	\$0.00
Amp	Fosi Audio v3 Stereo Amplifier 300W 48V power supply	1	\$116.99	\$116.99
Dual TRS to Dual RCA	J&D Dual 1/4 inch TS to Dual RCA Cable, 6 Feet	1	\$14.56	\$14.56
Speaker Wire	50ft Spool High Performance 16 Gauge Speaker Wire, Oxygen Free Copper	1	\$23.25	\$23.25
Banana Plugs	6-pairs 24K gold plated banana plug solders compression style	1	\$15.98	\$15.98
Diagnostics				
Measurement Mic	UMM-6 USB Measurement Microphone	1	\$94.98	\$94.98

Signal Chain & Diagnostics Subtotal

\$266

Speaker Components

External Components		Quantity	Item Cost	Total Cost
MDF	4'x8' 3/4" MDF	2	\$49.98	\$99.96
Banana Jack	Square Gold Binding Post Banana Jack 3-1/8"	2	\$14.58	\$29.16
Tweeter	Dayton Audio ND28F-6 1-1/8" Soft Dome Neodymium Tweeter 6 ohm	2	\$31.96	\$63.92
Woofer	Dayton Audio RS150P-8A 6" Reference Paper Cone Woofer 8 ohm	2	\$109.96	\$219.92
Port	Precision port 2" flared speaker cabinet port tube kit. Extendable	2	\$28.10	\$56.20
Internal Components				
Crossover Board	Black perforated Large Hole Crossover board pair 5" by 7"	1	\$8.29	\$8.29
0.8mH Inductor	Jantzen Audio 0.8mH 18AWG Air Core Inductor	2	\$24.62	\$49.24
0.9mH Inductor	Jantzen Audio 0.9mH 18AWG Air Core Inductor	2	\$24.50	\$49.00
0.2mH Inductor	Dayton Audio 0.2mH 20 AWG Air Core Inductor Crossover Coil	2	\$7.18	\$14.36
0.4mH Inductor	Dayton Audio 0.4mH 20 AWG Air Core Inductor Crossover Coil	2	\$8.78	\$17.56
10 Ohm Resistor	Jantzen Audio 10 Ohm 10 Watt Resistor	2	\$12.46	\$24.92
4.1 uF Capacitor	Jantzen 4.1 uF 400V Cross Cap Capacitor	2	\$4.62	\$9.24
2.4uF Capacitor	Jantzen 2.4 uF 400V Cross Cap Capacitor	2	\$3.34	\$6.68
10uF Capacitor	Dayton Audio DMPC-10 10uF 250V Capacitor	2	\$12.36	\$24.72

5.6uF Capacitor	Dayton Audio DMPC-5.6 5.6 uF 250V Capacitor	2	\$3.78	\$7.56
18 AWG Wire	Consolidated Stranded 18 AWG Hook-up wire	1	\$6.98	\$6.98
22 AWG Wire	22 AWG Silicone Electrical Wire Cable 6 coors 10 ft each	1	\$8.98	\$8.98
Acoustic Foam	12 Pack 1.5"x12"x12" sound proofing egg crate foam pad	1	\$12.99	\$12.99
Finishing		Quantity	Item Cost	Total Cost
Primer	Rust-Oleum Painter's Touch 2X Ultra Cover Spray Primer Flat Black	2	\$5.98	\$11.96
Spray Paint	Rust-Oleum Painter's Touch 2X Ultra Cover Spray Paint, Flat Black	2	\$6.98	\$13.96
Natural Wood Filler	Gorilla All Purpose Wood Filler, Natural Color, 6 Ounce Tube	1	\$8.68	\$8.68
Assembly		Quantity	Item Cost	Total Cost
Solder	MG Chemicals 60/40 Rosin Core Leaded Solder 0.032" Diameter 0.6 oz	1	\$10.94	\$10.94
Wood Glue	Titebond 5003 II Premium Wood Glue, 8-Ounces	1	\$5.44	\$5.44
Hot Glue	Owned	1	\$0.00	\$0.00
Screws	Owned Misc to use to fasten Woofer and Tweeter	1	\$0.00	\$0.00
Jigsaw	Borrowed	1	\$0.00	\$0.00
Drill + Bits	Owned	1	\$0.00	\$0.00
Sandpaper	Owned 60/120/220/320/400 grit assortment	1	\$0.00	\$0.00
Finishing Sander	Owned	1	\$0.00	\$0.00
Soldering Iron	Owned	1	\$0.00	\$0.00
Multimeter	Owned	1	\$0.00	\$0.00
Bar Clamps	Owned 4 x 8"	1	\$0.00	\$0.00
Hot Glue Gun	Owned	1	\$0.00	\$0.00
Table Saw	Borrowed	1	\$0.00	\$0.00
Modeling Software		Quantity	Item Cost	Total Cost
WinISD	Free Software	1	\$0.00	\$0.00
VituixCAD	Free Software	1	\$0.00	\$0.00
RoomEQWizzard (REW)	Free Software	1	\$0.00	\$0.00
Sketchup	Free Online	1	\$0.00	\$0.00
Speaker Components Subtotal				\$761
Project Total				\$1,026